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Compact XOR/XNOR-Based Adders and BNNs Utilizing Drain-Erase Scheme in Ferroelectric FETs

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ABSTRACT Compact and energy-efficient computing avenues such as in-memory computing and processing-in-memory (PIM) are being actively explored to address the limitations of the sparse von-Neumann computing systems. The recent advancements in the field of emerging non-volatile memories (e-NVMs), such as FeFETs, RRAMs, MRAMs, etc., have propelled the development of the PIM technique where the logic operations are performed in situ (where the operands are stored) to reduce the energy draining data movement. Considering the promising potential of the doped-hafnium oxide (HfO_2) based FeFETs, such as CMOS compatibility, high scalability, high integration density, and field-driven programming capability, in this work, for the first time, we propose a novel input-to-voltage mapping scheme and exploit drain-erase phenomenon to realize compact and energy-efficient majority logic gate using a single Fe-FDSOI FET, XOR and XNOR logic gates using two Fe-FDSOI FETs. Furthermore, utilizing the proposed FeFET-based XOR and XNOR logic design, we demonstrate a compact implementation of a half adder (using 3 FeFETs) and full adder (utilizing only 9 FETs) which outperforms the CMOS and prior eNVM-based implementations in terms of area and energy. Moreover, we also propose a modified XNOR-cell design utilizing 4 FeFETs for performing bitwise count operations in binary neural networks.

INDEX TERMS Drain-erase, ferroelectric FETs, full adder, processing-in-memory, XOR logic.

I. INTRODUCTION

The unprecedented growth in the field of processing-in-memory (PIM) architectures is attributed to the rapid advancements in the emerging non-volatile memories (e-NVMs) such as resistive random-access memory (RRAM), phase change memory (PCM), ferroelectric (Fe)FETs, etc. [1], [2]. Recently, CMOS-compatible doped-hafnium oxide-based FeFETs have garnered significant attention for application in memory and non-von-Neumann computing platforms owing to their high program/erase speeds, high scalability, large integration density, and multi-bit capability [3], [4], [5].

Considering the potential benefits of the FeFETs for PIM implementation such as high scalability, high endurance, run-time reconfigurability, low program/erase energy etc., in this work, for the first time, we propose a novel input-to-voltage mapping scheme (input assignments) for compact and energy-efficient implementation of XOR, XNOR and 3-input majority logic gates by exploiting the drain erase scheme in Fe-FDSOI FETs. The implementation of a 3-input majority gate was initially reported in [7]. Furthermore, the proposed FeFET-based XOR and XNOR logic gates serve as efficient building blocks for highly compact half-adder and full adder implementations, which require the least number

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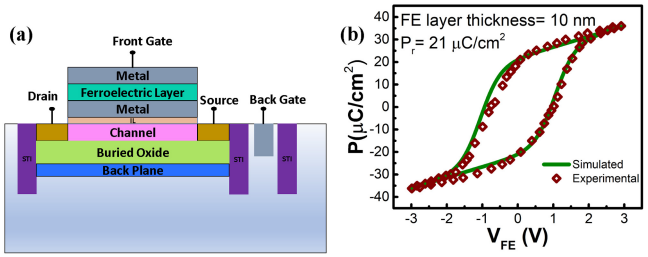


FIGURE 1. **(a) Schematic of the Fe-FDSOI FET having W = 170 μm and L = 24 μm. A 10 nm thick doped HfO_2 layer is present in the front-gate stack. (b) Polarization-voltage characteristics of FE-CAP model calibrated to the experimental results [6].**

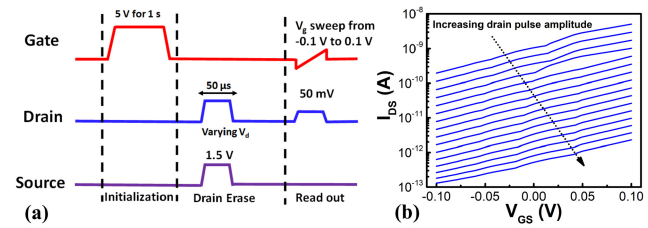


FIGURE 2. **Drain erase scheme: (a) voltage waveforms for drain erase scheme. (b) Read characteristics for different drain erase pulses.**

of transistors when compared to the conventional CMOS-design and the prior implementations [8]. Moreover, we also propose a novel 4-FeFET based XNOR cell to perform the computationally intensive multiply and accumulate (MAC) operations in binary neural networks using bitwise counting (XNOR+POPcount) while eliminating the additional weight fetching step required by the prior XNOR-BNNs.

II. FE-FDSOI SIMULATION METHODOLOGY

The structure of a Fe-FDSOI FET utilizing doped-hafnium oxide as the ferroelectric layer in the gate stack is shown in Fig. 1(a). For emulating the Fe-FDSOI FET, we have utilized a multi-domain Preisach model [9] for the FE capacitor connected to the experimentally calibrated industry standard BSIM IMG model [10]. The model parameters of the FE-Cap model are fine-tuned to reproduce the experimental characteristics [6] (Fig. 1(b)). While the input voltage ramp rate for the experimental data [6] is 24 mV/μs, we have used a ramp rate of 6 mV/μs for the model.

Moreover, it was shown recently that the drain terminal can also be used to alter the polarization state of the Fe layer in the gate stack, enabling the vertical stacking of FeFET cells in a 3D NAND architecture [11]. Fig. 2 illustrates this drain-erase operation: the FeFET is first initialized to a low V_{TH} state by applying a large positive voltage pulse at the gate terminal, as shown in Fig. 2(a). After initialization, voltage pulses of different amplitudes are applied to the drain terminal, with the source terminal kept at 1.5 V to enhance the impact of drain-erase [11]. Fig. 2(b) shows transfer (read) characteristics of the FeFET corresponding to the different drain-erase voltage pulses. From Fig. 2(b), it can be observed

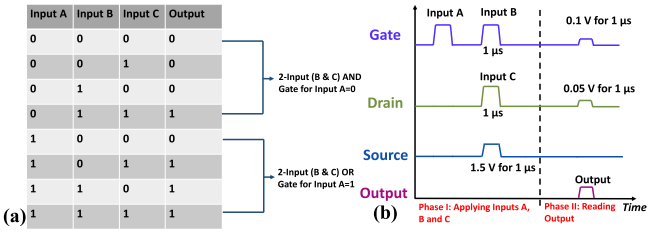


FIGURE 3. **(a) Truth Table of a 3-Input majority gate.(b) Voltage waveforms for the proposed 3-Input majority gate (here Input case A = 1, B = 1, and C = 0 is shown).**

that the drain current of the FeFET reduces as the amplitude of the drain-erase pulse is increased.

III. IMPLEMENTATION OF THREE-INPUT MAJORITY GATE

For the implementation of the three-input majority gate, we have utilized the ability of Fe-FDSOI FETs to program/erase (PRG/ERS) via the application of appropriate voltage pulses at the gate and drain terminals. The truth table of the three-input (A, B, and C) majority gate is shown in Fig. 3(a). Looking at the truth table, we find that when Input A is at logic level '0', the majority gate function can be realized as an AND operation between inputs B and C, and when input A is at logic level '1' the function can be realized as an OR operation between inputs B and C. This can be written in an equation form as

$$MAJ(A,B,C) = \begin{cases} AND(B,C) & InputA = 0 \\ OR(B,C) & InputA = 1 \end{cases} \quad (1)$$

For the implementation of Equation (1), we have proposed a novel input-to-voltage mapping scheme and exploited the drain-erase phenomenon. First, input A needs to be encoded appropriately to ensure that the Fe-FDSOI FET gets tuned to AND-gate mode when A = 0 or OR-gate mode when A = 1. To realize this functionality, input A is encoded as logic level '0' by applying a -5 V pulse of 1 μs duration at the gate terminal, whereas it is encoded as logic level '1' by applying a 5 V pulse of 1 μs duration at the gate terminal. On the other hand, input B is encoded as logic level '0' by applying a 0 V pulse for 1 μs at the gate terminal and logic level '1' by applying a 5 V pulse of 1 μs duration at the gate terminal of the Fe-FDSOI FET. Furthermore, input C is encoded as logic level '0' by applying a 5 V pulse at the drain terminal and is encoded as logic level '1' by applying a 0 V pulse for 1 μs duration (negative logic). At first, input A is applied; after that, inputs B and C are simultaneously applied at the gate and drain terminal, respectively.

During the application of inputs B and C, the source terminal of the Fe-FDSOI FET is kept at 1.5 V to increase the effectiveness of the drain-erase scheme [11]. After the application of inputs, the state of the Fe-FDSOI FET represents the output of the majority gate. The output is encoded as the current flowing through the Fe-FDSOI FET. The output state is read by applying a read gate voltage of 0.1 V and a drain voltage of 0.05 V. The high drain current

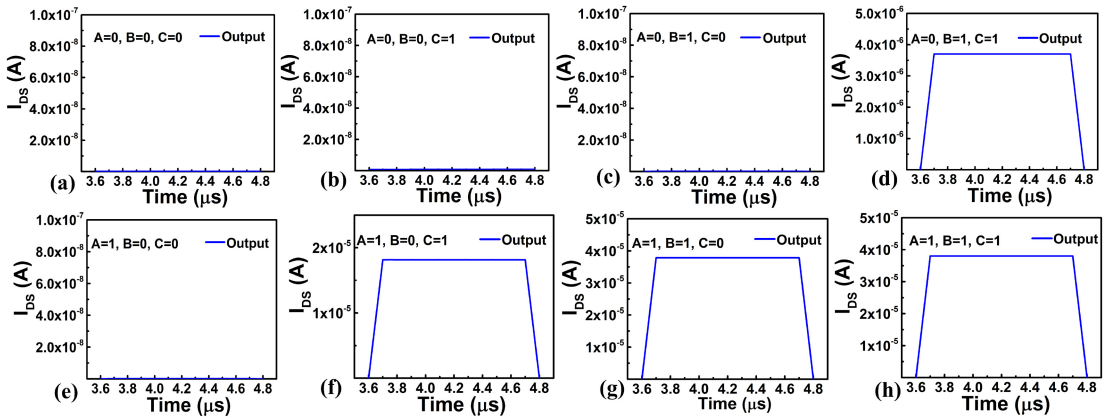
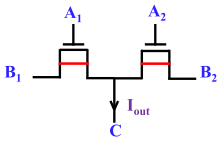


FIGURE 4. Output of the proposed majority gate for all possible input combinations are shown in (a)-(h). Sufficient separation between the output current levels corresponding to logic '0' and logic '1' (> 10^4) is observed.

Input A	Input B	Output C
0	0	0
0	1	1
1	0	1
1	1	0

(a)



(b)

	A_1	A_2
A=0	0 V	5 V
A=1	5 V	0 V
	B_1	B_2
B=0	0 V	5 V
B=1	5 V	0 V

(c)

FIGURE 5. (a) The truth table and (b) the proposed circuit for the XOR gate. Input A is divided into A_1 and A_2, while Input B is divided into B_1 and B_2. (c) Input-to-voltage mapping of the proposed XOR circuit.

flowing through the Fe-FDSOI FET represents logic level '1', and the low drain current represents logic level '0'. The operation can be understood as input A = 0 (-5 V initial pulse) erases the Fe-FDSOI FET inducing a high V_{TH} (low-current) state, and only for the case B = 1 (5 V pulse) and C = 1(0 V pulse), the Fe-FDSOI FET gets programmed and enters into a low V_{TH} state (high current, output = 1). Moreover, A = 1 (5 V pulse) also leads to the programming of the device in a low V_{TH} state (high current), and only for the case B = 0 (0 V pulse) and C = 0 (5 V pulse) the drain-erase scheme kicks in and the device switches to a high V_{TH} state (low current, output = 0). Thus, the device performs an AND operation between inputs B and C when A= 0 and performs an OR operation between inputs B and C when A = 1.

Fig. 3(b) shows the voltage waveforms of the entire operation for a representative input case. The output characteristics of the proposed three-input majority gate for all possible combinations of inputs are shown in Fig. 4(a)-(h). The maximum and the average read energy required to perform the logic operation is 1.9 pJ and 0.577 pJ, respectively.

IV. XOR/XNOR IMPLEMENTATION

The truth table of a two-input XOR gate is shown in Fig. 5(a). We propose a novel input-to-voltage mapping scheme (Fig. 5(c)) where the inputs A and B are divided into sub-inputs A_1-A_2 and B_1-B_2, respectively, to implement the XOR gate using two Fe-FDSOI FETs. While the sub-inputs A_1 and A_2 are applied to the gate terminals of the Fe-FDSOI FETs, the sub-inputs B_1 and B_2 are applied to the drain terminals as shown in Fig. 5(b). A logic '0' for input A is encoded by applying pulses of 0 V and 5 V amplitude with a duration of $1\mu s$ at the gate terminals A_1 and A_2, respectively, while logic level '1' is encoded by applying pulses with 5 V and 0 V amplitude and duration of $1\mu s$ at A_1 and A_2, respectively. Similarly, the logic '0' for input B is encoded by applying pulses of 0 V and 5 V amplitude with a duration of $1\mu s$ at the drain terminals B_1 and B_2, respectively, while logic level '1' is encoded by applying pulses with 5 V and 0 V amplitude and a duration of $1\mu s$ at B_1 and B_2, respectively. For reading the output, pulses with amplitude of 0.1 V and 0.05 V are applied at the gate and drain terminals, respectively, while keeping the source terminal grounded. The output C is encoded as logic '0' or '1' depending on the current flowing through the source terminal common to the two Fe-FDSOI FETs. When the inputs are different (A = 0/1 and B = 1/0), the novel input-to-voltage mapping scheme leads to the application of a 5 V pulse to the gate terminal of one of the Fe-FDSOI FETs while keeping its drain terminal grounded. This results in programming of the Fe-FDSOI FET to the low V_{TH} state and a consequent high current through the source terminal leading to output logic '1'. On the other hand, when the inputs are same (A = 0/1 and B = 0/1), both Fe-FDSOI FETs maintain their initial reset-state since 0 V is applied to the gate and drain terminals of one Fe-FFDSOI FET while 5 V is applied to the gate and drain terminals of the other FE-FDSOI FET. The voltage waveforms for the proposed XOR implementation for all possible input combinations are shown in Figs. 6(a)-(h).

The truth table of a two-input XNOR gate is shown in Fig. 7(a). We propose a similar input-to-voltage mapping scheme for FeFET-based XNOR gate implementation where inputs A and B are divided into A_1-A_2 and B_1-B_2, respectively, as shown in Fig. 7(b). The entire operation of the proposed XNOR gate is similar to the proposed XOR gate

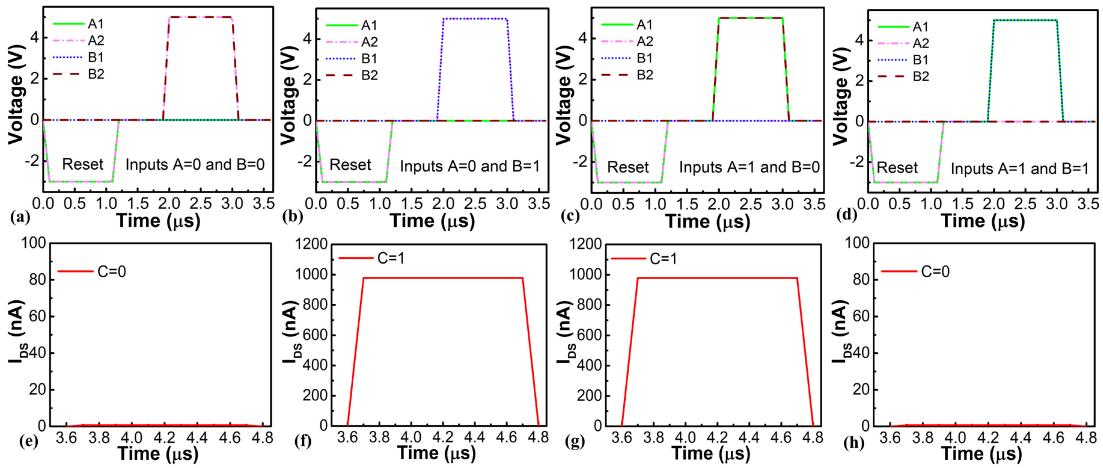
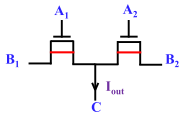


FIGURE 6. Input-to-voltage mapping scheme for the proposed XOR circuit for all possible input combinations are shown in (a)-(d) and their corresponding output waveforms are shown in (e)-(h).

Input A	Input B	Output C
0	0	1
0	1	0
1	0	0
1	1	1

(a)



(b)

	A_1	A_2
A=0	5 V	0 V
A=1	0 V	5 V
	B_1	B_2
B=0	0 V	5 V
B=1	5 V	0 V

(c)

FIGURE 7. (a) The truth table and (b) the proposed circuit for the XNOR gate. Input A is divided into A_1 and A_2, while Input B is divided into B_1 and B_2. (c) Input-to-voltage mapping for the proposed XNOR circuit.

with only a difference of input to voltage encodings in A_1-A_2.

The voltage and output current waveforms for different input combinations are shown in Figs. 8(a)-(e). For implementing the proposed input-to-voltage mapping, an additional inverter is required for providing complementary inputs which increases the overall transistor count. Moreover, while the inputs are encoded in the voltage domain in the proposed logic design, the output is encoded in the current domain. Therefore, for driving the next stage, i.e., for cascading, the output current levels of the preceding stage need to be converted to appropriate voltage levels before providing it as input to the next stage. Hence, the proposed implementation necessitates peripheral read-out circuits that utilize sense amplifiers similar to the prior processing-in-memory implementations utilizing emerging non-volatile memories [12], [13], [14]. To achieve this, a clocked pull-up p-type FET may be connected between the supply and output node [8] or a simple read-out circuitry using an inverting amplifier and a buffer can be used [15]. Furthermore, the readout circuitry can be shared among various logic functions in array architecture leading to a more efficient design while maintaining appreciable performance levels [2], [15]. Furthermore, all the logic-in-memory operations are performed in a pulsed-manner by providing the inputs as pulses at the gate/drain terminals with $1\mu s$ duration and a rise and fall time of 100 ns. The application of inputs perturbs the polarization-state of the FeFET which is measured by sensing the output drain current. The energy consumption of the proposed FeFET-based XOR and XNOR gate implementations are also evaluated for different input combinations in Table 1. The FeFET-based XOR/XNOR gates consume a significantly low (worst case) energy of 2.33 nJ.

We have used a gate and drain voltage of $V_g = V_d = 5$ V in our proof-of-concept demonstration to ensure a large separation between the output (drain) current values corresponding to logic '0' and logic '1' while keeping the output drain current corresponding to logic '1' at an appreciable level which may be easily detected by a simple sense amplifier. These voltage levels can be reduced significantly by using dual-port FeFETs where the read operation can be performed using the back gate [16] or using novel design techniques such as the use of Ge-channel [17]. Moreover, programming/erase voltages above 5 V have been utilized in scaled FeFETs for realizing a high-speed operation and a large memory window [18], [19]. Furthermore, all the logic-in-memory operations are performed in a pulsed-manner by providing the inputs as pulses at the gate/drain terminals with $1\mu s$ duration and a rise and fall time of 100 ns. The application of inputs perturbs the polarization-state of the FeFET which is measured by sensing the output drain current.

V. ADDER IMPLEMENTATIONS

A. HALF ADDER

The truth table for a half-adder is shown in Fig. 9(a) and the corresponding implementations of the sum (S= A⊕B) and carry bit (C_o = A.B) utilizing the Fe-FDSOI FETs are shown in Fig. 9(b). The proposed half adder implementation consists of three Fe-FDSOI FETs: while two Fe-FDSOI FETs are required to compute the sum (XOR operation), the third Fe-FDSOI FET is used to compute the carry bit by

TABLE 1. Read and write energy of two-input XOR and XNOR gate for different input combinations.

Logic	Inputs A=0 and B=0			Inputs A=0 and B=1			Inputs A=1 and B=0			Inputs A=1 and B=1		
	Write Energy	Read Energy	Total Energy	Write Energy	Read Energy	Total Energy	Write Energy	Read Energy	Total Energy	Write Energy	Read Energy	Total Energy
XOR	2.33 nJ	31.9 aJ	2.33 nJ	0.213 nJ	48.5 fJ	0.213 nJ	0.213 nJ	48.5 nJ	0.213 nJ	2.33 nJ	31.9 aJ	2.33 nJ
XNOR	0.213 nJ	48.5 fJ	0.213 nJ	2.33 nJ	31.9 aJ	2.33nJ	2.33 nJ	31.9 aJ	2.33nJ	0.213 nJ	48.5 fJ	0.213 nJ

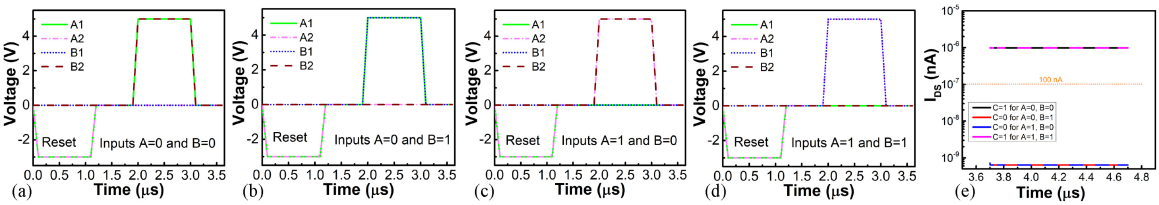


FIGURE 8. Input-to-voltage mapping for the proposed XNOR circuit for all possible input combinations are shown in (a)-(d). (e) Output waveforms with current in log scale for XNOR gate with 100 nA as reference.

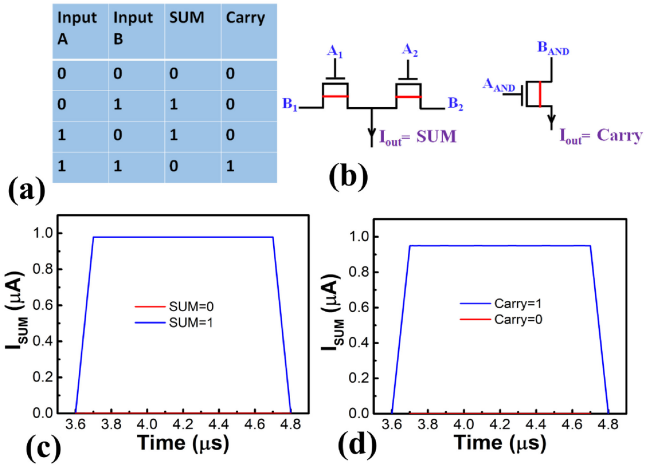


FIGURE 9. (a) Truth table and (b) proposed circuit for sum and carry operation in the half adder. (c) Input voltage and (d) output current waveforms for all combination of inputs for the proposed half adder.

performing AND operation (parallelly). While the operation of the Fe-FDSOI FET-based XOR gate has been discussed in Section III, the AND gate operation can also be performed utilizing a single Fe-FDSOI FET exploiting the drain-erase scheme with a novel input-to-voltage mapping scheme. For AND gate operation, input A is encoded as logic level '0' ('1') by applying a pulse of magnitude 0 V (5 V) with $1\mu s$ duration at the gate terminal while input B is encoded as logic level '0' ('1') by applying a pulse of magnitude 5 V (0 V) and $1\mu s$ duration at the drain terminal. The drain current represents the output which is read by applying a gate voltage of 0.1 V and a drain voltage of 0.05 V. The novel input-to-voltage mapping scheme for the AND gate ensures that the Fe-FDSOI FET gets programmed to the low V_{TH} state (and exhibits a high drain current) only for input case A = B = 1 (5 V at gate and 0 V at drain terminal) and shows negligible drain current for other combination of inputs. Fig. 9(c) and Fig. 9(d) show the output waveforms for the sum and the carry block, respectively, for different input combinations. The proposed half-adder implementation consumes a maximum energy of 96 fJ during the operation.

B. FULL ADDER

The truth table for a 1-bit full adder is shown in Fig. 10(a). The general expression for the sum of a 1-bit full adder can be given as:

$$\begin{aligned} SUM &= C_i \oplus (A \oplus B) \\ &= C_i\left(A'B' + AB\right) + C_i'\left(A'B + AB'\right) \end{aligned} \tag{2}$$

Utilizing the proposed FeFET-based XOR and XNOR gates, we propose a compact two-phase implementation of the SUM expression which consists of four Fe-FDSOI FETs and two FDSOI FETs as shown in Fig. 10(b). While the four Fe-FDSOI FETs are required to perform the XOR and XNOR operation between the inputs A and B, the outputs of the XOR and XNOR block are applied to the drain terminal of two FDSOI FETs M_1 and M_2 for AND operation with inputs C_i' and C_i, respectively. The input C_i is divided into C_{i1} and C_{i2} and encoded as logic level '0' ('1') by applying a 1 V (0 V) and 0 V (1 V) pulse of duration $1\mu s$ at the gate terminals of M_1 (representing C_{i1}) and M_2 (representing C_{i2}), respectively, as shown in Fig. 10(c). The entire process of computing the SUM bit is divided into two phases: the results of the XOR and XNOR operations are obtained in the first phase while the input C_i is applied as C_{i1} and C_{i2} and the output is read in the second phase as shown in Fig. 11.

Now, the carry operation of the 1-bit full adder represented by:

$$C_{out} = AB' + BC_i' + A'C_i \tag{3}$$

can be performed by utilizing the proposed AND gate implementation with three Fe-FDSOI FETs as shown in Fig. 12(a). The input-to-voltage encoding scheme for the carry operation are listed in Fig. 12(b). Similar to the SUM operation, the carry computation is also divided into two phases: the inputs A, B, and C_i are applied in phase I for AND operation while read biases are applied in the second phase to obtain the carry bit. The output waveforms for different input combinations are shown in Fig. 13. The sum and carry bit of the proposed 1-bit full adder are computed at the same time. The total energy consumption for the

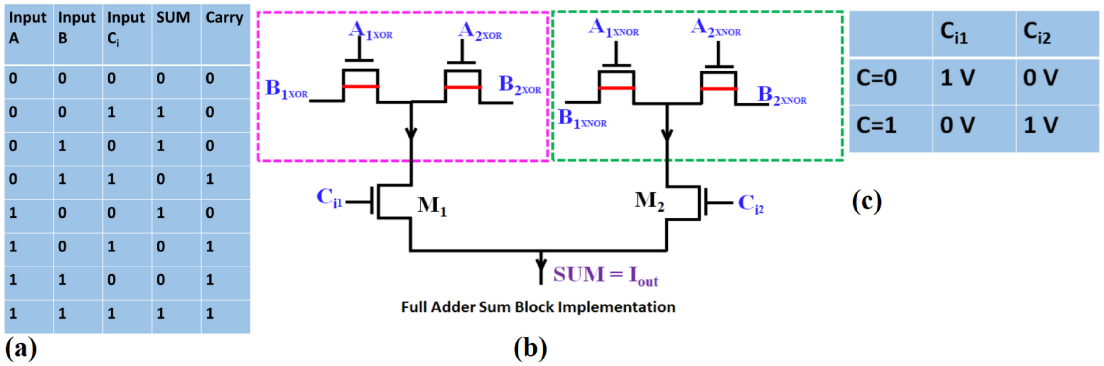


FIGURE 10. (a) Truth table and (b) proposed circuit for the full adder sum block. XOR and XNOR blocks used in full adder sum block are marked by pink and green dashed boundaries. (c) The input-to-voltage mapping scheme for C_i input.

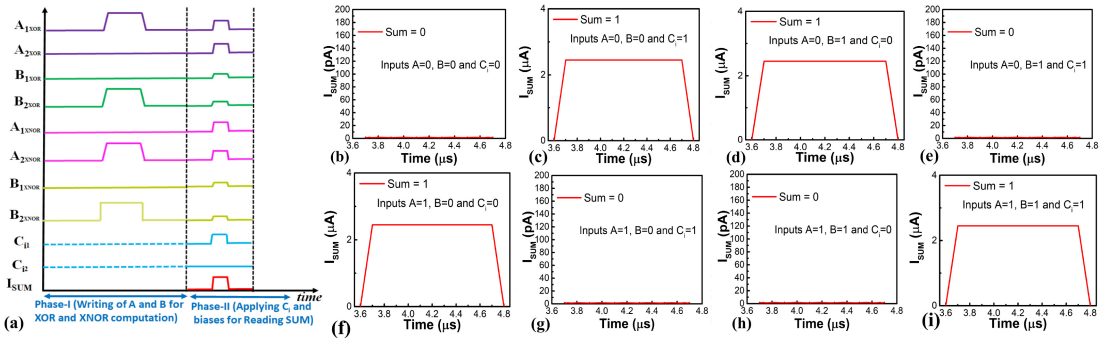


FIGURE 11. (a) Representation of voltage waveforms of two phases for input A = 1, B = 0 and C_i=0 for the proposed full adder sum block. Output current waveforms for all possible input combinations are shown in (b)-(i).

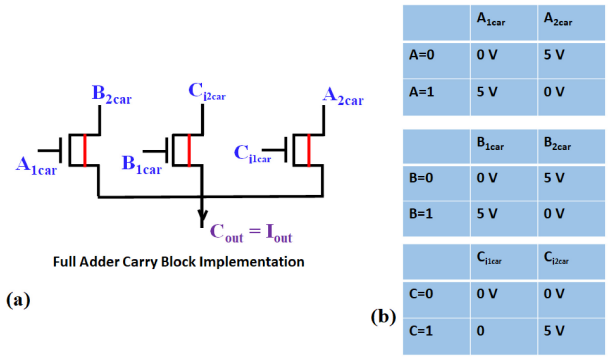


FIGURE 12. (a) Circuit for carry block of the proposed full adder and (b) the input-to-voltage encodings utilized for its implementation.

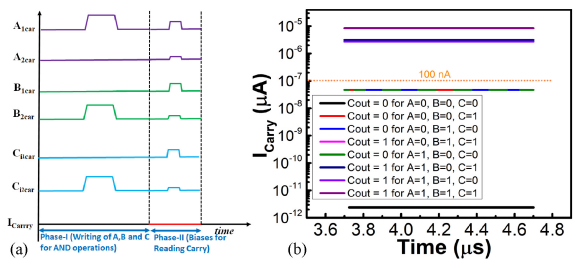


FIGURE 13. (a) Representation of input voltage waveforms for input A = 1 B = 0 and C_i=0 for the full adder carry block. Output waveforms with current in log scale for carry operation of the full-adder are shown in (b).

proposed full Adder for all possible input cases are reported in Table 2. As can be observed from Fig. 8(e), while the output of the XNOR gate shows consistent currents for logic '0' and logic '1', the current levels for output logic '0' and logic '1' for the carry operation in the full adder implementation is different for different input combinations as shown in Fig. 13(b). Therefore, to ensure consistency in the proposed logic design, we use 100 nA as the reference current level for the readout circuitry and classify the output current level above 100 nA as logic '1' and output current less than 100 nA as logic '0'. Moreover, we have benchmarked the performance of the proposed full adder against the recent stand-alone PIM implementations based on eNVMs in Table 3. The proposed FeFET-based full adder implementation requires only 9 transistors and exhibits the least footprint as compared to the prior implementations. The proposed logic design methodology can be extended to logic-in-memory array architectures with shared readout circuitry such as [20]. The output of the logic functions encoded as the polarization-state of the FeFETs can be sensed using readout circuitry and then applied as next set of inputs to the array. Moreover, logic operations with a common input can be performed simultaneously by applying the common input to the word lines (gate terminals) while applying

TABLE 2. Read energy consumption of 1-bit full adder for different input combinations.

Inputs	Read Energy for SuM	Read Energy for Carry	Total Read Energy
000	0.086 aJ	0.1195 aJ	0.2055 aJ
001	122.5 fJ	2.34 fJ	124.84 fJ
010	122.5 fJ	2.34 fJ	124.84 fJ
011	0.086 aJ	156 fJ	156 fJ
100	122.5 fJ	2.34 fJ	124.84 fJ
101	0.086 aJ	156 fJ	156 fJ
110	0.086 aJ	156 fJ	156 fJ
111	122.5 aJ	400 fJ	522.5 fJ

the other inputs to different bit lines (drain terminals) in an AND-array architecture. Extensive array-level design analysis and exploration of the proposed FeFET-based logic-in-memory implementation is an important future work. Similar to the prior works on FeFETs [8], [21], our in-house developed compact model also exhibits a switching time of 1 μs. Furthermore, we have used rise/fall times of 100 ns, input/read pulse widths of 1 μs, an initial reset pulse duration of 1 μs and applied the inputs 1 μs after the application of the reset pulse. Also, the read operation for determining the output of the logic gate is performed 500 ns after the application of the input pulses to account for the read-after-write delay in FeFETs caused by parasitic charge trapping [22]. Therefore, the proof-of-concept demonstration of the proposed FeFET-based logic-in-memory implementation takes 4.8 μs. However, many recent works have used a pulse width less than 300 ns to switch between the extreme polarization-states in FeFETs [18], [23]. Furthermore, recently, a switching time of 300 ps for a pulse amplitude of 4.5 V was shown experimentally utilizing proper impedance matching technique and dedicated RF probes [24]. It was also reported that the switching time of 300 ps is limited by the measurement capabilities, and the minimum switching time should be 1-10 ps based on the nucleation propagation theory [24]. Moreover, by using SiNx as an interfacial layer, which results in negligible charge trapping [25] and also by applying an appropriate standby bias [26], the read-after-write delay can be substantially reduced to 10 ns. Therefore, the latency of the proposed FeFET-based logic-in-memory implementation can be reduced significantly.

VI. XNOR-BITCOUNT BASED BNN ACCELERATOR

A. IMPLEMENTATION

Although the deep neural networks (DNNs) and convolutional neural networks (CNNs) show unprecedented accuracy surpassing the human capabilities for tasks such as image classification, natural language processing, and speech recognition [27], they are compute- and resource-intensive owing to the huge number of multiply and accumulate operations (MAC) [28]. This restricts their deployment on the resource-constrained IoT edge devices with conventional von-Neumann computing engines. To reduce the computational complexity of the DNNs and CNNs and enable

TABLE 3. Performance benchmarking of 1-bit full adder.

Device Technology	No. of devices used	Read Energy (or power consumed)
FeFET [8]	20 FETs (7FeFETs and 14 FETs)	avg: 1.42 fJ min: 0.42 fJ
FeFET [32]	32 FETs (28 FETs and 4 FETs)	0.54 fJ (0.27 μW)
FeFET [32]	20 FETs (17 FETs and 3 FeFETs)	0.42 fJ (0.21 μW)
STT+SHE[1] [33]	52 FETs and 4 MTJs	1.23 fJ
SHE-MTJ[2] [34]	23 FETs and 3 SHEs	40.68 fJ ($\sim$13.6 μW)
Memristor [35]	9 memristors[3]	6.4 nJ
CMOS* [36]	$>$ 14 FETs	60.6 μW
Dynamic CMOS* [37]	16 FETs	2.14 μW
Reconfigurable FETs[4] [38]	8 RFETs	NA
This Work[5]	9 FETs (7 FeFETs and 2 FETs)	avg: 170.62 fJ (0.17 μW) min: $<$aJ

[1]STT: Spin-transfer torque, SHE: Spin-Hall effect
[2]MTJ: Magnetic Tunnel Junction
[3] Requires 43 steps
[4] Additional gates needed to drive the transmission gates are neglected
[5] Inverter count not included
* Overhead of data transfer not included

their deployment on IoT edge devices with limited area and energy, network quantization techniques where weights and activations are restricted to two (binary) values were proposed recently [29], [30]. This quantization technique in the binary neural networks facilitates simplified execution of the computationally demanding multiply and accumulate (MAC) operations via bitwise counting (which involves XNOR+POPcount) [30]. These XNOR-bitcount based BNNs exhibit a significant reduction in the computational workload and the memory for storing parameters without compromising significantly with the accuracy. The energy-efficiency and the performance of these BNNs may be further improved by utilizing in-memory computing architecture based on eNVMs specifically tailored for XNOR Neural Networks (XNNs). To this end, in this work, we also propose a FeFET-based XNOR cell for compact and energy-efficient BNN accelerators. The compact and energy-efficient FeFET-based XNOR gate implementation can be used to perform the XNOR operation of the weights and inputs. However, the polarization-state of the Fe-FDSOI FETs changes according to the XNOR output and the information regarding the weights is lost. Therefore, the weights need to be fetched again for the XNOR operation with the next set of inputs during the inference leading to additional energy-hungry data movements. Although the weights can be stored in a SRAM cell and fetched whenever required for XNOR operation similar to [31], the transistor count increases considerably for one cell of the XNN array degrading the area-efficiency of the BNN accelerator.

To address this bottleneck, we have proposed a modified structure of XNOR cell specifically tailored for XNNs as shown in Fig. 14(a). It consists of four Fe-FDSOI FETs where two (computing) Fe-FDSOI FETs (C1 and C2) are used for performing XNOR operation while the remaining two (memory) Fe-FDSOI FETs (M_1 and M_2) are used to store weights and eliminate the weight fetching step and the related energy-hungry data movements. First, the binary

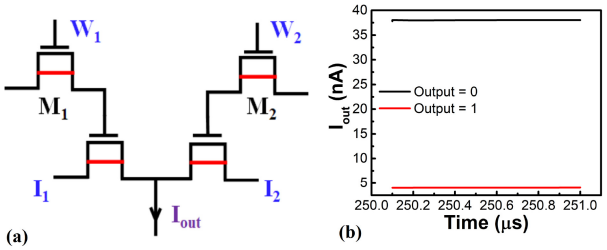


FIGURE 14. (a) Schematic of the circuit for XNOR cell in XNN. (b) Read characteristics of XNOR cell.

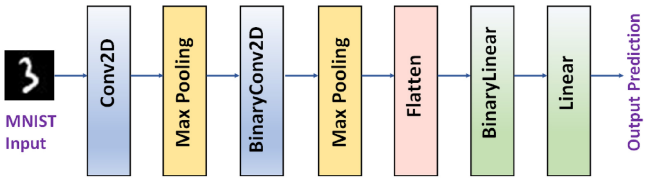


FIGURE 15. Architecture of neural network used in network-level simulations.

weight is programmed as the polarization-state of M_1 and M_2. To store weight '0', a positive programming pulse of duration 50 μs is applied at the gate terminal of M_1, and the gate terminal of M_2 is grounded. For storing weight '1', the gate terminal of M_1 is grounded, and a positive programming pulse of duration 50 μs is applied at the gate terminal of M_2. Now for performing the XNOR operation, the input is divided into two sub-inputs: I_1 and I_2 and the logic '0' ('1') for input I is encoded by applying pulses of 0 V (5 V) and 5 V (0 V) amplitude with a duration of $1\mu s$ at the drain terminals I_1 and I_2, respectively. The result of the XNOR operation is stored as the polarization state of the FE layer in the gate stack of C1 and C2 while the weights are preserved in the memory Fe-FDSOI FETs (M_1 and M_2) for inference operation with next set of inputs. For obtaining the output, appropriate read voltages are applied to all four Fe-FDSOI FETs. Fig. 14(b) shows the read characteristics of the proposed XNOR cell of XNN, where the output drain current is high when both the inputs are same while the output drain current is low when the inputs are different.

B. IMPACT OF WEIGHT FLIPPING

Ferroelectric FETs are also known to exhibit unique spatial (device-to-device) variation due to the process non-uniformity. Therefore, application of the same programming pulse to different FeFETs in the proposed XNOR-cell array may result in different polarization-states. As a result, some of the binary weights may not be correctly programmed to their desired polarization-state resulting in flipped weights during the inference operation in XNNs. We have performed a comprehensive analysis of the impact of weight flipping in the proposed FeFET-based XNN by carrying out extensive network level simulations.

The architecture used to perform the network-level simulations of the proposed XNOR neural network is shown in Fig. 15. This architecture is an adaptation of

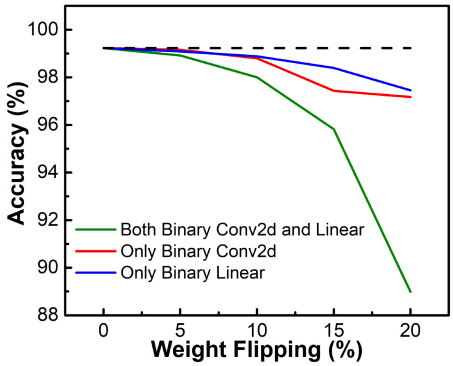


FIGURE 16. Impact of weight flipping in binary Conv2d and Linear layers on accuracy.

the XNOR-Net [30] integrated into a modified LeNet5 framework. It consists of two convolutional layers (one binary) and two linear layers (one binary). The MAC operations in the binary layers were performed using bitwise count operations (XNOR+POPcount) for binary weights and activations [30]. Network-level simulations of the proposed XNOR-based BNNs without weight flipping indicates that a 99.21% accuracy can be achieved on the MNIST dataset. Fig. 16 shows the impact of weight flipping in the binary convolutional and linear layers on the accuracy. Flipping of weights in both binary convolutional and binary linear layers leads to a significant degradation in the accuracy of the network. The accuracy degrades to 89% when 20% of the weights in both binary convolutional and binary linear layers are randomly flipped. We have also analyzed the sensitivity of the network accuracy to the weight flipping in the binary convolutional layer and binary linear layer separately. The network accuracy is more sensitive to the weight flipping in the convolutional layer as compared to the weight flipping in the linear layer, as shown in Fig. 16. Also, weight flipping in individual layers (Linear or Conv2d) only does not degrade the network accuracy significantly.

VII. CONCLUSION

In this work, utilizing novel input-to-voltage mapping schemes, we have proposed compact implementations of logic circuits such as 3-input majority gate (using one FeFET), XOR and XNOR gates (using only two FeFETs), half adder (using only 3 FeFETs) and full adder (using only 9 FETs) which outperform the prior implementations in terms of area and energy. We have also proposed a novel FeFET-based XNOR-based cell that eliminates the weight fetching step and the related energy-hungry data movements while performing bitwise count operation in BNNs. The impact of weight flipping on the accuracy of image classification of the MNIST data set was also evaluated in the proposed FeFET-based XNOR Neural Network. We believe that our promising results may provide the incentive for the experimental realization of area and energy-efficient in-memory Boolean logic circuits using ferroelectric FETs in this nascent era of computing.

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